Future-Context Requirements for Streaming Accent Conversion: A Controlled Study Using Causal Phone Translation

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***Abstract*—Streaming foreign accent conversion (FAC) systems choose a lookahead budget and report a single end-to-end latency, but to our knowledge prior streaming FAC work has not densely swept that budget while separately reporting algorithmic and computational latency. We train a causal phone translator at 16 lookaheads from 0 to 640 ms with every other factor held fixed, and report four results. First, quality improves smoothly with lookahead at -0.045 phone error rate (PER) per doubling ($R^2 = 0.98$) with *no locatable knee*: a piecewise fit is preferred under every axis treatment we tried, yet the breakpoint moves between 40 and 180 ms and no bootstrap interval is narrower than three octaves. Instead of a knee we identify *saturation* at ≈ 340 ms: with the grid completed to 20 ms spacing over 200–360 ms, and against a repeatability floor measured from deliberate fixed-seed repeats ($2\sigma = 0.0029$), every observed 20 ms step beyond 340 ms buys less than that floor. The figure ranges 300–480 ms across the floor's confidence interval. At the 40 ms lookahead used by PHONOS's translator, our probe retains a further 39.8% relative PER reduction between 40 and 240 ms. Second, a pre-registered hypothesis that vowels and approximants degrade faster than obstruents is *not supported*: all seven manner classes improve by 52–72% relative, and the between-group difference is smaller than the dispersion among classes within the groups. Bootstrapping over the six test speakers rather than over phone tokens, that difference is $+0.040$ with a 95% interval of $[-0.010, +0.101]$. It still fails when errors are reweighted by measured acoustic cost, so this is not an artefact of PER charging every error equally. Third, canonical-phone conversion benefits more from future context than accent-faithful transcription at matched capacity (-0.045 against -0.029 PER per doubling, $1.55\times$), and within the conversion arm the benefit is $1.32\times$ larger at positions where the speaker actually deviated from canonical. Fourth, per-chunk compute depends only weakly on lookahead — at most 1.9% on an Apple M4 and 7.6% on an ARM64 Linux host across a $17\times$ change in algorithmic latency — empirically decoupling lookahead from per-chunk compute over the tested range, subject to the throughput constraint $t_{\mathrm{cmp}} < c$. We release the harness and the per-condition outputs.**

***Index Terms*—accent conversion, streaming speech, latency, lookahead, self-supervised models**

I. Introduction

Streaming FAC is established. PHONOS [2] runs at ≤ 40 ms lookahead and reports an 81% reduction in non-native accent confidence. What does not exist is any account of *why* 40 ms, what it costs in quality, or what any of it does on hardware that is not a datacentre GPU.

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Harness, causality proof, sweep configurations, analysis scripts with self-tests and per-condition hypothesis dumps: https://github.com/Dipeshtripathi13/streaming-fac-lookahead. Data is L2-ARCTIC [1] via `KoelLabs/L2Arctic` (CC-BY-NC-4.0); that licence is non-commercial, so configurations are released rather than weights.

Three gaps follow, and this paper addresses the first two directly.

The budget is chosen, not characterised. PHONOS *states* 40 ms and publishes no ablation showing 40 ms is necessary or sufficient; TVTSyn fixes 4 future tokens ($\approx$80 ms) without varying them. DarkStream does ablate, over three points $\{0, 140, 280\}$ ms, and finds the third buys <1pp — but three points scored on anonymisation accuracy cannot locate where returns stop, which is the question a deployment budget actually asks. "What would 160 ms buy you" still has no published answer.

Algorithmic and computational latency are reported fused. A figure such as "<241 ms on a single GPU" combines a modelling choice with unstated inference and buffering costs. A reader cannot tell whether a faster chip would help. We decompose throughout as

$$t_{\mathrm{e2e}} = \underbrace{c + L}_{t_{\mathrm{alg}}} + t_{\mathrm{cmp}} + t_{\mathrm{buf}}, \tag{1}$$

where c is chunk size, L is lookahead, t_{cmp} is per-chunk compute and t_{buf} is I/O and jitter buffering. t_{alg} is inherent: it is the delay on an infinitely fast machine, because the model refuses to emit frame n until it has seen frame $n{+}L$. t_{cmp} is implementation- and hardware-dependent and can be reduced through optimization or faster hardware.

Degradation is reported in aggregate. Existing results are utterance-level MOS, WER or accentedness. We found no prior streaming FAC study reporting lookahead sensitivity by phonetic manner class.

Our contributions:

1) A latency–accuracy characterization of a causal phone-translation probe for streaming FAC across 16 lookahead budgets, reporting an average exchange rate and a floor-relative saturation region rather than an unidentifiable knee (§V-C).
2) A negative result on a pre-registered phoneme-class hypothesis (§V-D).
3) Evidence that canonical-phone prediction benefits more strongly from lookahead than realized-phone transcription under matched capacity, and that this holds within a single arm after controlling for speaker and phone identity (§V-E).

4) Two patches that make a masked SSL encoder causal, with truncation-based causality verification (§III-B).
5) Three documented measurement failures, each of which produced plausible output before being caught, and a measured repeatability floor that supersedes an earlier over-conservative estimate (§VI).

II. Related work

Streaming FAC is established. Nguyen et al. [3] present a streaming non-autoregressive accent-conversion system on an Emformer encoder, demonstrating feasibility at stable latency; PHONOS [2] subsequently targets much lower latency with bounded future context. Neither characterises how quality varies with that context, which is the axis we isolate. PHONOS, TVTSyn [4] and DarkStream [5] come from one group and pair conversion with speaker anonymisation, and two report more than is sometimes credited. TVTSyn gives CPU as well as GPU latency ($\approx$132 and $\approx$79 ms at 60 ms chunk), so a CPU number is not unpublished — a CPU latency–*quality* curve is. DarkStream ablates look-ahead over $\{0, 140, 280\}$ ms and selects 140 ms because the extra 140 ms buys $<$1pp at double the buffering delay: a three-point ablation, scored on anonymisation accuracy rather than conversion quality, whose marginal return collapses in the same region where we place saturation on phone error rate over 16 budgets. That agreement across objectives is a stronger position than absent prior evidence would have been.

StreamVC [6] reports mobile operation without device, core or thread details, limiting hardware-level reproducibility. A survey [7] frames the space; streaming anonymisation [8], [9], [10] shares the constraint but not the accent objective; offline FAC [11], [12] and low-latency VC [13] bracket the design space. Our contribution is orthogonal: we characterise an axis these systems parameterise but do not study, and claim no quality improvement over any of them.

III. Method

A. A causal phone translator as the probe

Training a full FAC pipeline per lookahead is prohibitive. We instead isolate a phonetic-normalization component directly relevant to the research question; future-context needs in acoustic generation and prosody may differ. L2-ARCTIC [1] provides, per utterance, both the canonical G2P/lexicon-derived phone sequence (`g2p`) and the annotated realized phone sequence (`ipa`). We operationalize the phonetic-normalization component of accent conversion as

non-native audio $\rightarrow$ *canonical phone sequence*,

a CTC head over a frozen causal encoder. This provides a controlled supervised proxy for the phonetic-normalization stage, and swapping the target tensor from `g2p` to `ipa` yields a more tightly matched architectural control, differing in *one tensor*, than an AC-versus-VC comparison — though the two targets still differ in label entropy, sequence length and annotation noise (§V-E).

TABLE I
TRUNCATION TEST ON `WAVLM-BASE-PLUS`. RELATIVE L_2 CHANGE IN EARLIER FRAMES AFTER DELETING LATER AUDIO.

configuration	relative L_2	causal?
attention mask only	1.14×10^{-2}	no
+ causal positional conv	6.00×10^{-3}	no
+ cumulative GroupNorm	2.60×10^{-2}	no
both patches	$\mathbf{6.14 \times 10^{-6}}$	**yes**

B. Making the encoder causal

Masking self-attention in wav2vec2/WavLM-base [14], [15] does *not* produce a causal encoder. Two leaks remain: the positional convolution (`pos_conv_embed`, depthwise, kernel 128, symmetric padding) admits 1.28 s of future at a 20 ms frame rate; and the feature-encoder GroupNorm normalises each channel over the entire utterance, making every output frame depend on every input frame.

We verify causality by truncation: delete audio after frame t and check whether any earlier frame moved (Table I). Note the third row: patching GroupNorm alone makes matters *worse*. A partial causality fix is not a partial improvement.

C. Lookahead is quantised

Lookahead is realized as $N_L = \lceil L/h \rceil$ frames at a frame hop $h = 20$ ms (equivalently, a 50 Hz frame rate). A knee narrower than 20 ms is not merely unmeasured but *unrepresentable*, and sampling finer than f silently creates duplicate conditions that reweight any fit. Lookahead is therefore quantized in 20 ms increments: we sample every increment through 200 ms and a subset of quantized budgets above it, and assert zero collisions in t_{alg} across all conditions before fitting.

IV. Experimental setup

Frozen causal WavLM-base-plus, CTC phone head, 1200 steps, batch 8, chunk 40 ms, 100-frame lookback, speaker-disjoint test split of 400 utterances. Two sweeps: a 7-lookahead $\times$ 2-target $\times$ 3-seed sweep (42 runs) and a 16-lookahead $\times$ 2-seed dense sweep, run on *both* target arms (64 runs), all on an NVIDIA T4. Compute measurements use an Apple M4 and an ARM64 Linux host, CUDA-event or `perf_counter_ns` timed, 28–50 repetitions with three warm-up iterations discarded.

Repeatability floor. A conventional multi-seed design does not isolate fixed-seed run-to-run nondeterminism: pairing within seed cancels seed effects but leaves this residual, which arises from nondeterministic GPU kernels. We measure it directly, repeating four conditions ($L \in \{40, 160, 240, 320\}$ ms) four times each at *identical seed and configuration*, giving 12 degrees of freedom on test PER:

$$\sigma = 0.00146, \qquad 2\sigma = \mathbf{0.0029}\text{ PER}, \tag{2}$$

with a bootstrap 95% interval of $[0.0013, 0.0034]$ on 2σ.

This supersedes an earlier estimate of $2\sigma = 0.0044$ taken from six *accidental* repeats, which the present interval excludes; that estimate was inflated by a single outlying pair.

TABLE II

CONVERGENCE CHECK: VALIDATION PER AGAINST OPTIMIZATION STEP, SINGLE SEED. THE RANKING IS CORRECT AT ALL 12 CHECKPOINTS AND THE FITTED EXCHANGE RATE IS STABLE FROM STEP 1000 ONWARD.

step	L=0	40	160	240	640
500	0.5301	0.4375	0.3288	0.2946	0.2574
1000	0.4715	0.3601	0.2588	0.2353	0.2012
2000	0.4480	0.3392	0.2332	0.2054	0.1717
3000	0.4340	0.3293	0.2131	0.1930	0.1622
4000	0.4135	0.3152	0.1992	0.1873	0.1533
5000	0.4092	0.3108	0.1962	0.1838	0.1473
6000	0.4060	0.3100	0.1951	0.1822	0.1468

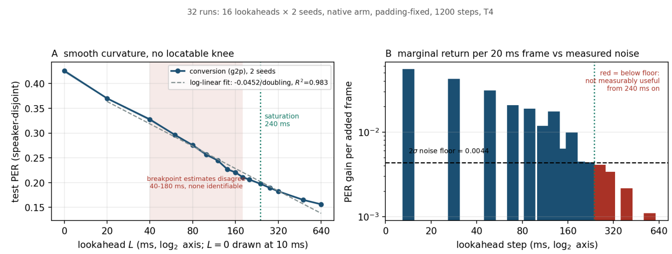

Fig. 1. Dense sweep, conversion arm, 16 lookaheads $\times$ 2 seeds. **Left:** smooth curvature; the shaded band spans the breakpoint estimates produced by three defensible treatments of L=0, none identifiable. **Right:** marginal return per added 20 ms frame against the measured 2σ floor; red bars are below it.

The correction matters because the saturation point in §V-C is defined against this floor, and a smaller floor makes smaller gains measurable. We also note the floor is not constant across conditions — per-condition σ spans 0.00064 to 0.00223, a factor of 3.5 — so a single global threshold is an approximation, and marginal decisions near it should be read as such.

V. RESULTS

A. Per-chunk compute depends only weakly on lookahead

Across $L \in [0, 640]$ ms at chunk 40 ms on the M4, t_{alg} spans 40–680 ms (17×) while median per-chunk compute moves from 43.23 ms to at most 44.06 ms — a range of -0.3% to $+1.9\%$. On ARM64 the span is 0 to $+7.6\%$. The two terms of Eq. 1 are therefore independently actionable, which is the practical justification for reporting them separately.

Independently actionable is not the same as free. At this configuration $t_{\text{cmp}} \approx$ 43–44 ms against a 40 ms chunk, i.e. $\text{RTF} = t_{\text{cmp}}/c \approx 1.08 > 1$: the base encoder does not sustain real time at 40 ms chunks on this machine, whatever L is. Eq. 1 describes the delay of a single chunk and must be read alongside the steady-state throughput constraint

$$\text{RTF} = t_{\text{cmp}}/c < 1, \tag{3}$$

which lookahead leaves untouched but chunk size and hardware do not. Increasing L adds little per-chunk compute over the tested range; being able to run at all is a separate constraint.

B. Is the curve an optimization-rate curve?

Every condition elsewhere is trained for a fixed 1200 steps, so what is measured directly is *lookahead → performance after 1200 optimization steps*. That equals *lookahead → attainable performance* only if the conditions sit at comparable points in optimization. If longer lookahead simply trained faster, part of the exchange rate would be an optimization-rate difference rather than a context effect. We test this rather than list it as a caveat.

Five representative budgets ($L \in \{0, 40, 160, 240, 640\}$ ms) were trained for 6000 steps — five times the standard budget — with validation every 500 steps, giving a 12-point trace per condition (Table II).

Three results. **Training largely flattens**: over the final 1500 steps the conditions move by 0.0005–0.0045 PER. Against the measured 0.0029 floor (Eq. 2) three of the five (L=40, 160, 240) have settled below it, while L=0 (0.0041) and L=640 (0.0045) are still drifting slightly above — so 6000 steps is near, but not fully at, convergence for the two extremes. **The ranking never reorders**: the ordering L=0 $>$ 40 $>$ 160 $>$ 240 $>$ 640 holds at **12 of 12** checkpoints, including the earliest at step 500. **The exchange rate is stable**: fitted over $L \geq 40$ it stays within -0.0396 to -0.0461 PER per doubling at every checkpoint, and the value at the 1200-step budget used elsewhere (≈ -0.0405) agrees with the 6000-step value (-0.0414) to within 0.001, far inside the noise floor.

Improvement from step 1500 to 6000 is also roughly uniform across conditions (-0.030 to -0.045 PER) rather than concentrated at long lookahead, which is the specific pattern a convergence-rate confound would produce. Absolute PER does keep falling with longer training, so the fixed-budget numbers elsewhere are a comparison at matched budget rather than converged performance — but the *relative* quantities this paper reports, the ranking and the exchange rate, are not artefacts of stopping at 1200 steps. The check is single-seed over five budgets and uses validation rather than test PER, so it bounds the concern rather than eliminating it.

C. RQ1: an exchange rate and a saturation point, not a knee

Conversion-arm PER falls monotonically from 0.4256 at L=0 to 0.1561 at L=640 ms. Fitting $L \geq$ 20 ms on $\log_2 L$ gives

$$\Delta\text{PER} = -0.0452 \text{ per doubling}, \quad R^2 = 0.983, \tag{4}$$

over 15 points.

There is curvature, and it is not localisable. BIC prefers a two-segment fit under all three axis treatments we tried (ΔBIC $= -22.2$, -31.2, -36.8), which by the usual thresholds [16] is strong evidence. But the breakpoint lands at 40, 180 and 180 ms respectively, and every bootstrap 90% interval spans about three octaves. Sixteen points did not rescue a knee. The reason is mechanical and worth stating: on a $\log_2(L{+}1)$ axis the L=0 → 20 ms gap is 4.39 units while every other gap is ≈1.0, so a piecewise fit places a breakpoint just past that gap whatever the data does.

What we report instead is the lookahead beyond which marginal return falls below the measured floor. Because the original grid was 20 ms-spaced only to 200 ms, we added the five budgets 220, 260, 300, 340 and 360 ms under identical

TABLE III
CONVERSION-ARM PER AGAINST LOOKAHEAD (MEAN OF 2 SEEDS) AND MARGINAL RETURN PER ADDED 20 MS FRAME, IN UNITS OF THE MEASURED 2σ=0.0029 FLOOR (EQ. 2). BELOW 1.0× AN ADDITIONAL FRAME IS NOT MEASURABLY USEFUL. THE GRID IS 20 MS-SPACED TO 360 MS; THE FINAL TWO ROWS AVERAGE OVER WIDER INTERVALS.

L (ms)	PER	gain/frame	× floor
0	0.4256	—	—
20	0.3700	0.0556	19.2
40	0.3275	0.0426	14.7
60	0.2963	0.0312	10.8
80	0.2754	0.0209	7.2
100	0.2564	0.0190	6.6
120	0.2446	0.0118	4.1
140	0.2269	0.0176	6.1
160	0.2206	0.0064	2.2
180	0.2106	0.0099	3.4
200	0.2062	0.0045	1.5
220	0.2030	0.0031	1.1
240	0.1973	0.0057	2.0
260	0.1934	0.0039	1.3
280	0.1891	0.0043	1.5
300	0.1855	0.0036	1.2
320	0.1823	0.0032	1.1
340	0.1790	0.0033	1.1
360	0.1770	0.0020	**0.7**
480	0.1649	0.0020	**0.7**
640	0.1561	0.0011	**0.4**

settings, making the range 200–360 ms fully 20 ms-spaced. The quantity below is therefore a per-*observed*-step gain, not an average over wider sampled intervals.

Gains run $+0.056$ (0→20 ms), $+0.021$ (60→80) and $+0.0045$ (180→200). Against the measured floor of 0.0029 (Eq. 2), every observed 20 ms step remains above it out to 340 ms — 1.1× (200→220), 2.0, 1.3, 1.5, 1.2, 1.1, 1.1 — and every step beyond falls below: 0.7× (340→360), and 0.7 and 0.4 for the two longer intervals to 640 ms. We therefore place **saturation at ≈340 ms**: the point beyond which no observed 20 ms increment buys a measurable gain.

This figure moves with the floor, and we report the sensitivity rather than a single number. Across the floor's bootstrap interval the saturation point ranges from 300 ms (at $2\sigma = 0.0034$) to 480 ms (at 0.0013). Under the superseded 0.0044 estimate it would have been 240 ms — so the earlier figure was an artefact of an over-conservative floor, not of the curve. The gains between 200 and 340 ms sit at only 1.1–2.0× the floor, so this is a region where returns become marginal rather than a sharp cutoff. Unlike a knee, though, the quantity depends on the floor rather than on an axis choice, and the floor is now measured with 12 degrees of freedom.

The cost of a 40 ms budget. PER at L=40 ms is 0.3275; at 340 ms it is 0.1790. **Relative to the 40 ms condition, increasing lookahead to the saturation point reduces PER by a further 45.3%**, and to 640 ms by 52.3%. Measured instead against the full range available here (L=0 → 240 ms), a 40 ms budget has already captured 57% of the attainable reduction and leaves 43% unclaimed. "Current budgets are under-provisioned" is thus supportable as diminishing-returns geometry with a measurable saturation point — not as a knee.

TABLE IV
RELATIVE ERROR-RATE REDUCTION, L=0 → 640 MS. B = PRE-REGISTERED AS BREAKING FIRST, S = AS SURVIVING. INTERVALS ARE PERCENTILE BOOTSTRAP OVER THE SIX TEST SPEAKERS, NOT OVER PHONE TOKENS. THE PREDICTED GROUPING DOES NOT ORDER THE DATA.

	class	rel. gain	95% CI	ref. phones
S	nasal	0.716	[0.668, 0.779]	4 020
B	approximant	0.700	[0.628, 0.774]	4 575
S	fricative	0.687	[0.608, 0.774]	7 659
B	vowel (mono)	0.673	[0.595, 0.767]	13 008
B	vowel (diph)	0.654	[0.577, 0.758]	4 824
S	stop	0.615	[0.550, 0.689]	7 071
S	affricate	0.525	[0.397, 0.633]	1 098

D. RQ2: the preregistered manner grouping does not predict lookahead benefit

We pre-registered, before any of this data existed, that vowels and approximants would break first. The prediction is fixed in the public repository at commit `c5066dc` (2 Aug 2026), seven days before the hypothesis dumps it is tested on were committed (`b66d0a5`, 9 Aug 2026), so the ordering is verifiable rather than asserted. The prediction was that vowels and approximants break first while stops, fricatives, affricates and nasals would survive short lookahead. The reasoning was that vocalic identity is carried by formant movement across a syllable nucleus that is tens to hundreds of milliseconds long — Peterson and Lehiste measured English syllable nuclei at roughly 80–250 ms [17], a range consistent with the vowel durations later reported over 139 talkers [18] — and that this movement is criterial rather than incidental, since even nominal monophthongs are identified better from a trajectory than from a steady-state slice [19], with glides and diphthongs differing from vowels in rate of formant change rather than in kind [20]. Obstruents were placed on the other side because stop place is signalled by brief formant transitions to a locus [21] and is recoverable from a short-time spectrum at the release [22], so a narrow window should suffice. Testing this on 42,255 unique reference-phone tokens evaluated across seven lookahead conditions (295,785 phone–condition observations), with each substitution or deletion charged to the class of the *reference* phone:

Group means differ in the predicted direction (0.676 vs 0.636), but that is the weakest available reading and four stronger checks fail. The effect size is 0.66: the between-group difference (0.040) is smaller than the pooled SD across classes (0.060). The surviving group's *internal* spread is 0.190 — 4.7× the between-group difference. And 5 of 12 pairwise orderings are violated, with nasals out-gaining all three breaks-first classes.

At the right level of clustering the difference is not distinguishable from zero. The 42,255 tokens per condition are not independent: they are nested in six speakers, and the claim at issue is about accented speech, for which the independent replicate is a talker rather than a phone. Resampling speakers rather than tokens (2000 draws, Table IV) puts the between-group difference at $+0.040$ with a 95% interval of $[-0.010, +0.101]$, which **includes zero**. Every individual class gain remains large and clearly non-zero — the intervals in Table IV sit between 0.40 and 0.78 — so lookahead unambiguously helps every class; what fails to separate from zero is the *grouping*. Token-level intervals would have been roughly an order of magnitude narrower and would have declared this difference significant, which is the error the clustering avoids. The same procedure applied to RQ3 (Sec. V-E) *does* exclude zero, so the null here is a property of this grouping rather than of an estimator too conservative to resolve anything at six speakers.

The pattern does not hold within L1 either, but that split cannot carry weight. Splitting by native language, the ratio of group gains straddles unity: Vietnamese 1.23, Korean 1.19, Arabic 1.08, Spanish 1.02, Hindi 0.99, Chinese 0.92; requiring both the gain ratio and the mean slope to move in the predicted direction, only 3 of 6 qualify (Spanish clears the gain criterion but not the slope one). A grouping that reverses across a third of the languages in the corpus is not capturing a property of manner class. The deeper limitation is that in the speaker-disjoint test split *each L1 is represented by exactly one speaker* (ABA, LXC, SVBI, HJK, EBVS, PNV), so L1 and talker are perfectly confounded: this table cannot separate a language effect from an individual one, and every "by-L1" row is equally readable as a single speaker's idiosyncrasy. We report it as a consistency check on the pooled result — the reversals are informative because they show the pooled direction is not robust to *any* disaggregation — and not as evidence about native language.

H2 is not supported. The finding is not that lookahead fails to help vowels — every class improves 52–72% relative, all above the floor. The finding is that manner class does not predict which sounds need right context.

Reweighting each error by a proxy for its acoustic cost does not rescue it. PER charges every error exactly 1.0, so the natural objection is that the grouping is sound and PER simply mis-weights it: a substituted vowel and a substituted stop cannot sound equally wrong. We test that directly.

Each utterance's canonical phone string is synthesised with a single fixed voice and force-aligned to its own synthesis using an espeak-IPA CTC model, giving a time span per reference phone. The hypothesis string is synthesised with the same voice and DTW-aligned to the reference, and the spectral distortion inside each reference phone's own span is charged to that phone's class. Localisation matters here: a VITS-class synthesiser is globally coupled, so changing one phone perturbs the whole utterance and a whole-utterance distortion cannot be attributed to a single phone. The per-phone span from forced alignment is what makes the measurement local at all.

Synthesis must also be deterministic. Piper defaults to noise_scale = 0.667 and noise_w = 0.8, under which the same phone string renders differently on each call: self-distortion is 244.8 against 277.6 for a genuine one-phone change, so roughly 88% of an apparent "cost of an error" would be synthesis noise. Both terms are zeroed, giving bitwise-identical renderings, and the analysis refuses to run otherwise.

The measure is validated before it is used, and the verdict is suppressed if validation fails: phones the sequence metric scores as incorrect carry $1.72\times$ the distortion of correct ones (86.8 versus 50.5), and mean distortion falls monotonically with lookahead (71.4 to 46.9, -34%).

TABLE V
H2 UNDER ACOUSTIC WEIGHTING: RELATIVE REDUCTION IN PER-PHONE DISTORTION, L=0 $\rightarrow$ 640 MS, 40 UTTERANCES, 1451 REFERENCE PHONES PER CONDITION. B = PRE-REGISTERED AS BREAKING FIRST, S = AS SURVIVING. THE SEQUENCE-METRIC COLUMN IS COMPUTED ON THE SAME UTTERANCES, SO THE TWO DIFFER ONLY IN WEIGHTING.

	class	acoustic	sequence	phones
B	approximant	0.386	0.825	153
B	vowel (mono)	0.379	0.712	448
S	nasal	0.376	0.784	128
S	fricative	0.372	0.745	272
B	vowel (diph)	0.321	0.667	166
S	stop	0.305	0.663	240
S	affricate	0.257	0.526	44
effect size (need $\geq$ 1.0)		0.756	0.607	
ordering violations		2/12	4/12	

The verdict is unchanged (Table V). Acoustic weighting does move in H2's favour — effect size $0.607 \rightarrow 0.756$ and ordering violations $4/12 \rightarrow 2/12$ against the sequence metric on the same utterances — but not far enough to clear the pre-registered threshold of 1.0, and the failure has the same shape as before: the "survives" group's internal spread (0.120) is $3.5\times$ the difference between the groups (0.034), and a nasal again out-gains a diphthong. So this particular equal-cost objection is not supported by the synthesized distortion analysis; it does not exclude weightings this proxy cannot see, notably trajectory errors at correctly identified phones.

One explanation does remain open, and it is not the one this test closes. Both signals here are TTS renderings of phone strings, so the synthesiser produces canonical formant trajectories for whatever string it is given. This measures the cost of getting a phone's *identity* wrong; it says nothing about a wrong trajectory for a phone whose identity is right. That still needs real converted audio against a native rendition (§VII). The sample is also far smaller than the sequence-level test — 1451 phones per condition against 295,785 — and is weighted accordingly.

E. RQ3: canonical-phone prediction benefits more from lookahead than realized-phone transcription

Identical architecture, capacity, data order, seed and step count; the two arms differ only in the target tensor. Endpoint gain L=0 $\rightarrow$ 640 ms is -0.2695 (63.2% relative) for conversion against -0.1893 (42.8%) for accent-faithful transcription, a paired difference of $+0.0802$ (sd = 0.0035, t = 39.7).

The dense grid replicates this, and sharpens it. Running the full 16-lookahead $\times$ 2-seed sweep on the transcription arm as well puts both arms on the same dense grid (Table VI). The endpoint gains reproduce the 7-point result almost exactly — 63.3% for conversion against 43.5% for transcription, against 63.2% and 42.8% before — on an independent, denser sample. The exchange rates separate cleanly:

conversion -0.0452 vs transcription -0.0292 PER per doubling,

$R^2 = 0.983$ and 0.984 respectively, so **conversion buys $\mathbf{1.55}\times$ as much per doubling of lookahead**.

The effect is clearer in the between-arm gap than in the absolute levels. At L=0 the two arms have similar PER (0.4256 vs 0.4427, a gap of 0.0171); by L=640 ms the gap is 0.0939, **5.5× wider**. It widens in 13 of 15 adjacent steps, and the transcription arm is worse at 16 of 16 lookaheads, every one by more than the 0.0029 floor.

The arms differ in their target tensor, which carries more than the task: label entropy, sequence length and annotation noise move with it. We therefore repeat the test *inside* the conversion arm, where none of those can differ. Splitting reference positions by whether the speaker actually deviated from canonical (g2p $\neq$ ipa), the exchange rate is -0.0520 PER per doubling at accent-changing positions against -0.0394 at unchanged ones, a factor of $\mathbf{1.32}$ from the same model on the same utterances. That converges with the $1.55\times$ between arms. One caveat belongs with it: in *relative* terms the ordering reverses (51% against 71%), because accent-changing positions start far harder (0.590 against 0.354) and the gap between the two widens with lookahead. Absolute PER decreases faster at canonical-deviation positions, although these positions begin substantially harder and show a smaller relative reduction.

The interaction survives two controls. Those positions are not a random subset: they differ in baseline difficulty, phone composition and distribution over speakers. Treating 42,255 phone tokens nested in six speakers as independent would manufacture significance, so we bootstrap over *speakers*: the slope difference is -0.0127 with a 95% CI of $[-0.0211, -0.0069]$, excluding zero, and deviation positions are steeper in 2000 of 2000 resamples. Recomputing the difference *within each reference phone* and pooling gives -0.0098, steeper in 26 of 35 phones — so phone composition accounts for roughly a quarter of the raw gap but not the effect. We report this as an association under matched conditions, not as evidence that `g2p≠ipa` positions are exclusively where accent resides: such positions also include pronunciation variants, reductions and annotation error.

Two secondary points fall out. Both arms saturate at the same ≈240 ms, and neither yields a locatable knee (breakpoints move to $[40, 240, 160]$ ms across axis treatments for transcription, against $[40, 180, 180]$ for conversion), so §V-C's structural conclusion is not an artefact of the conversion arm. And the transcription arm's own curve is just as log-linear, so the shallower slope is a smaller effect of lookahead, not a noisier measurement of the same one.

The claim is carried by the *preference margin* — PER against the other arm's labels minus PER against its own — which grows monotonically in 3/3 seeds for conversion (+0.032 at L=0 to +0.099 at 640 ms; 18/18 adjacent step × seed comparisons positive) while the transcription control is 0/3 monotone and drifts *down* (-0.0103). Building the claim on a difference rather than a level turned out to matter (§VI).

TABLE VI
BOTH ARMS ON THE DENSE GRID, MEAN OF 2 SEEDS. THE GAP IS THE TRANSCRIPTION ARM'S PER MINUS THE CONVERSION ARM'S: NEAR ZERO AT L=0 AND WIDENING MONOTONICALLY WITH LOOKAHEAD.

L (ms)	conversion	transcription	gap
0	0.4256	0.4427	+0.0171
20	0.3700	0.3896	+0.0196
40	0.3275	0.3600	+0.0325
80	0.2754	0.3271	+0.0517
160	0.2206	0.2908	+0.0702
240	0.1973	0.2755	+0.0782
320	0.1823	0.2666	+0.0843
640	0.1561	0.2501	+0.0939
exchange rate (PER/doubling)			
conversion	-0.0452	$R^2 = 0.983$	
transcription	-0.0292	$R^2 = 0.984$	

F. t_{buf}: jitter measured, I/O not

t_{buf} splits into a jitter term and an I/O term. The first is measured; the second resisted three separate attempts, and we report that rather than substitute a number.

Jitter is negligible. Two independent 30 s duplex runs on the M4 at 40 ms blocks give required jitter buffers of 0.14 and 0.08 ms, p99 callback lateness ≈0.06 ms, zero over- or underruns, and clock drift below 1.3×10^{-4}%. On this platform the jitter safety margin contributes nothing meaningful to the budget.

The I/O term is unmeasured, and the reason is instructive. An acoustic loopback — emit a chirp, record it, cross-correlate — failed on every attempt, with all 20 repetitions rejected at correlation 0.042–0.046 against a 0.20 threshold. The diagnosis is not weak signal: captured level is *invariant* to output amplitude (peak 0.00505 at amplitude 0.50 versus 0.00468 at 0.95), so nearly doubling the output changed the captured signal by nothing. That is the signature of platform echo cancellation, which scales with its own reference and therefore cannot be defeated by a louder probe.

The driver-reported figure is not a substitute. Its excess over the requested block is *identically* 201.69 ms at both 10 and 20 ms blocks before roughly doubling at 40 ms, which identifies it as queueing rather than converter delay; real applications on this platform achieve sub-20 ms round-trip.

A digital loopback through a virtual audio device — the signal never becomes sound, so echo cancellation cannot reach it — also fails, for a different and more interesting reason. The path works mechanically, but the returned signal peaks near 0.014; lowering the detection threshold until repetitions pass yields p50 206 ms, p90 491 ms, $\sigma = 117$ ms. A distribution whose p90 is $2.4\times$ its p50 is not a latency distribution, and the honest description is that a threshold was lowered until the tool produced output. Such a device would also measure its own buffering rather than a converter. Closing this term needs a wired loopback through real converters, or a platform without echo cancellation in the path.

VI. MEASUREMENT AUDITS AND FAILURE MODES

A padding bug rewrote the curve's shape. A masked block handed a zero-padded hidden-state tensor to a depthwise convolution and feed-forward that do not respect the key-padding mask. The correction was not a constant offset — it grows with lookahead, from -0.014 PER at $L{=}0$ to -0.086 at 320 ms, $16.4\times$ the noise floor — so the buggy curve *understated* the benefit of lookahead, biasing the central quantity of RQ1. A constant offset would have cancelled in every within-run comparison; this did not. Every number reported in this paper was regenerated after the fix.

A breakpoint estimator returns a breakpoint whether or not one exists. ΔBIC cannot distinguish "there is a knee" from "piecewise fits a curve better than a line". Identifiability has to be tested by perturbing the axis and bootstrapping the location, not inferred from ΔBIC alone. Our synthetic check reproduces the artefact exactly: on a curve that is log-linear for $L \geq 20$ with $L{=}0$ off the extrapolation, including $L{=}0$ yields $\Delta\mathrm{BIC} = -39$ and a confident "knee at 40 ms", while dropping it yields $+5$ and no knee at all. An earlier maximum-distance-to-chord estimator reported "knee at 160 ms, bootstrap 95% CI $[160, 160]$" on this data — a tight interval around a plausible value, tight precisely *because* the curve is smooth and every resample of a log-linear curve is also log-linear.

An analysis that buckets unmatched symbols into "other" will produce confident output on the wrong alphabet. Our phoneme-class map was keyed on ARPAbet; the sweep emits IPA. Applying one to the other sent 57% of reference tokens — including *every* vowel and /ɹ/ — to "other", and still printed per-class slopes, an H2 verdict and a by-L1 table. Nothing errored. The guard is to assert coverage rather than infer it from the absence of an exception, which the analysis now does.

A fourth attempt, a TTS-counterfactual substitute for forced alignment, was built and rejected before producing any reported number; it is documented in the repository.

VII. Limitations

Every quality result here is phone-level, not perceptual, and this is a phone-translation probe rather than an end-to-end FAC system: the latency–accuracy relation reported here need not transfer to synthesis, prosody, speaker similarity or perceived accentedness. We do not conduct a perceptual study because the phone-sequence outputs are not valid FAC stimuli: synthesized from bare phone strings, they lack the lexical, prosodic and acoustic structure required to judge naturalness or accentedness. This is a property of the stimuli rather than of the budget. Training is 1200 steps at fixed budget over a frozen encoder — a comparison at matched budget rather than at convergence, though §V-B shows the ranking and exchange rate hold to $5\times$ that budget; the dense sweep has two seeds per arm and no repeated cells, so its noise floor is imported from six accidental repeats in the other sweep and carries no confidence interval of its own. t_{buf}'s I/O term is unmeasured, and compute is characterised on Apple Silicon and ARM64 Linux only.

VIII. Conclusion

For causal phone translation, future context buys accuracy at an approximately log-linear average rate of ≈ 0.045 PER per doubling, with no identifiable knee and measurable returns to saturation at ≈ 340 ms — well beyond PHONOS's 40 ms translator budget, relative to which a further 39.8% reduction is available. Lookahead changes algorithmic latency rather than per-chunk cost, though throughput must still satisfy $t_{\mathrm{cmp}} < c$ (Eq. 3) to run at all.

IX. Acknowledgements

This work received no external funding and used no institutional compute: all GPU results are from free-tier Google Colab (Tesla T4) and all CPU results from consumer hardware. Thanks to the maintainers of L2-ARCTIC, WavLM, sherpa-onnx and Piper, whose public releases made a single-author study of this scope possible.

X. Compliance with ethical standards

This work used only the publicly released L2-ARCTIC corpus and involved no new data collection and no human subjects, so no ethical approval was required. No listening study was run, and no claim about perceived accentedness or naturalness is made. The author declares no competing interests.

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